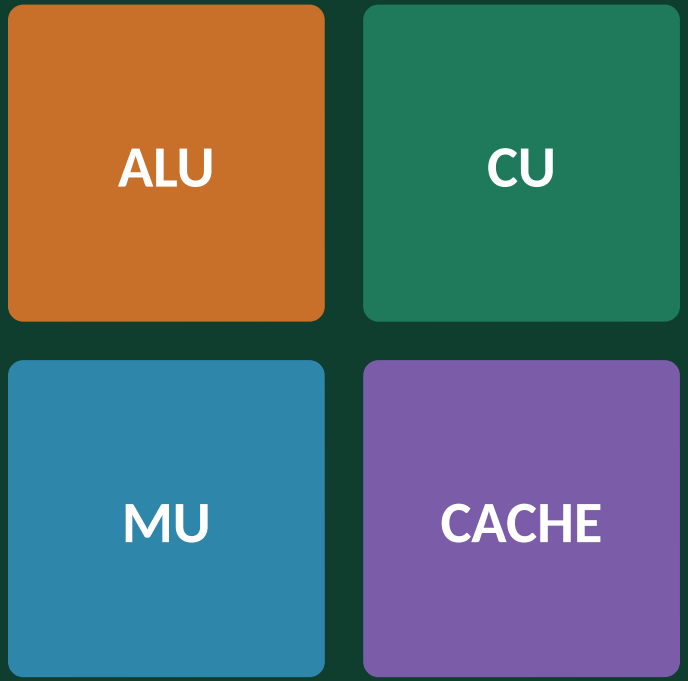
CPU

Central Processing Unit – the brain of the computer. Functions, components, generations, cores, buses and sockets explained in depth.



Main Important Points of CPU

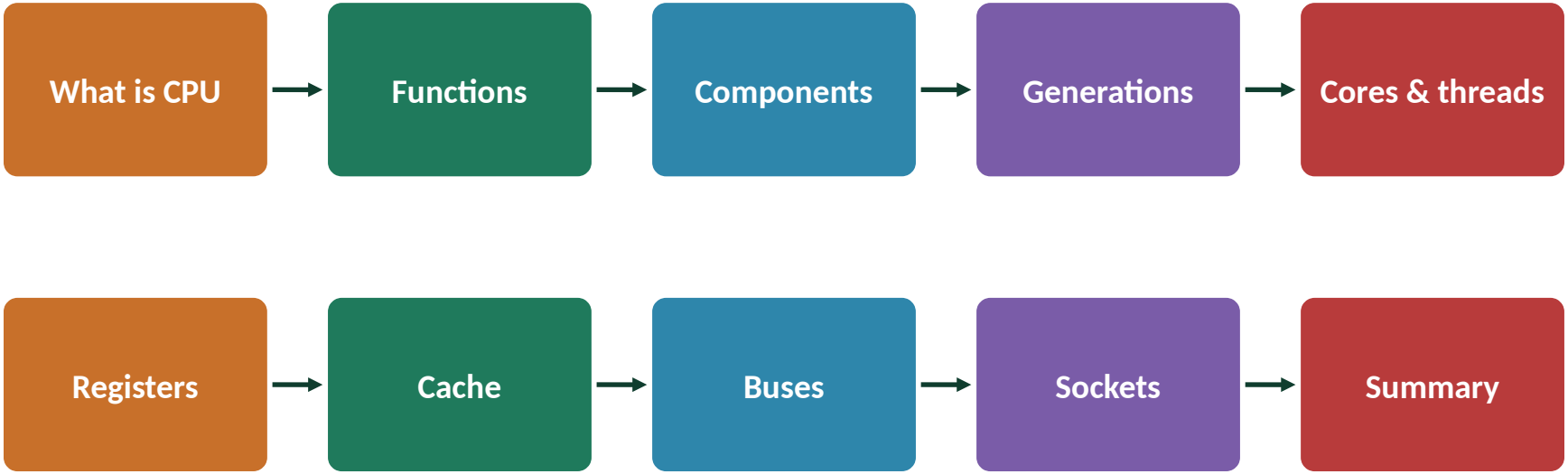
FROM THE LECTURE

- What is CPU?
- Functions of CPU
- Types of Microprocessor
- Components of CPU
- ALU, CU, MU, Cache Memory

FROM THE LECTURE

- Register Set (IR, AR, MA)
- Generation of CPU
- Core Technologies
- Hyper threading
- Internal Buses

Roadmap



Follow the topics in this order. Every topic is first shown as in the lecture, then explained step by step.

Microprocessor

FROM THE LECTURE

A microprocessor is **a multipurpose, programmable logic device** that reads binary instructions from a storage device called memory accepts binary data as input and processes data according to those instructions and provides result as output.

Microprocessor – Explained

1 Multipurpose

Can run any kind of work: games, documents, calculations, internet.

2 Programmable

Its behaviour is controlled by instructions (programs), so changing the program changes the job.

3 Logic device

Built from logic gates that make decisions with 0 and 1.

4 Input → process → output

Reads binary instructions from memory, accepts binary data, processes it and gives the result.

What is CPU?

FROM THE LECTURE

- It is known as the brain of the Computer. It is first priority to run Machine
- It is a chip based structure made up of millions of transistors and logic gates
- It receives the incoming instructions as input from Memory or RAM, Proceed it and gives result as Output
- It is Responsible for the Speed and Capacity to execute tasks in computer systems
- These Microprocessor are Categorized into 2 different architecture 32 BIT and 64 BIT

What is CPU? – Explained

1 Brain of the computer

Nothing runs without it. It decides what happens and in what order.

2 Chip of transistors

A transistor is a tiny electronic switch. Millions (today billions) of them form logic gates.

3 Input from RAM

Instructions and data are loaded from RAM, processed, and the result is sent out.

4 Speed and capacity

The CPU decides how fast tasks run and how much work the computer can handle.

5 32 or 64 BIT

The architecture tells how many bits the CPU handles at once.

6 Analogy

The CPU is the chef: RAM is the table with ingredients, the disk is the pantry.

Transistors and logic gates

Transistor

A microscopic switch: ON = 1, OFF = 0. Millions of them sit on one chip.

Logic gate

A few transistors combine into a gate such as AND, OR or NOT.

Circuits

Gates combine to add numbers, compare values and store bits.

Result

All the work of a CPU is billions of tiny ON/OFF decisions per second.

32 BIT vs 64 BIT

	32 BIT	64 BIT
Data handled at once	32 bits	64 bits
Max RAM used	About 4 GB	Hundreds of GB and far more
Operating system	32-bit Windows	64-bit Windows (also runs 32-bit apps)
Speed on big tasks	Lower	Higher
Today	Old systems	Standard in modern PCs

What decides CPU speed?

1 Clock speed

Measured in GHz: how many cycles per second. Higher means more work per second.

2 Number of cores

More cores can run more tasks at the same time.

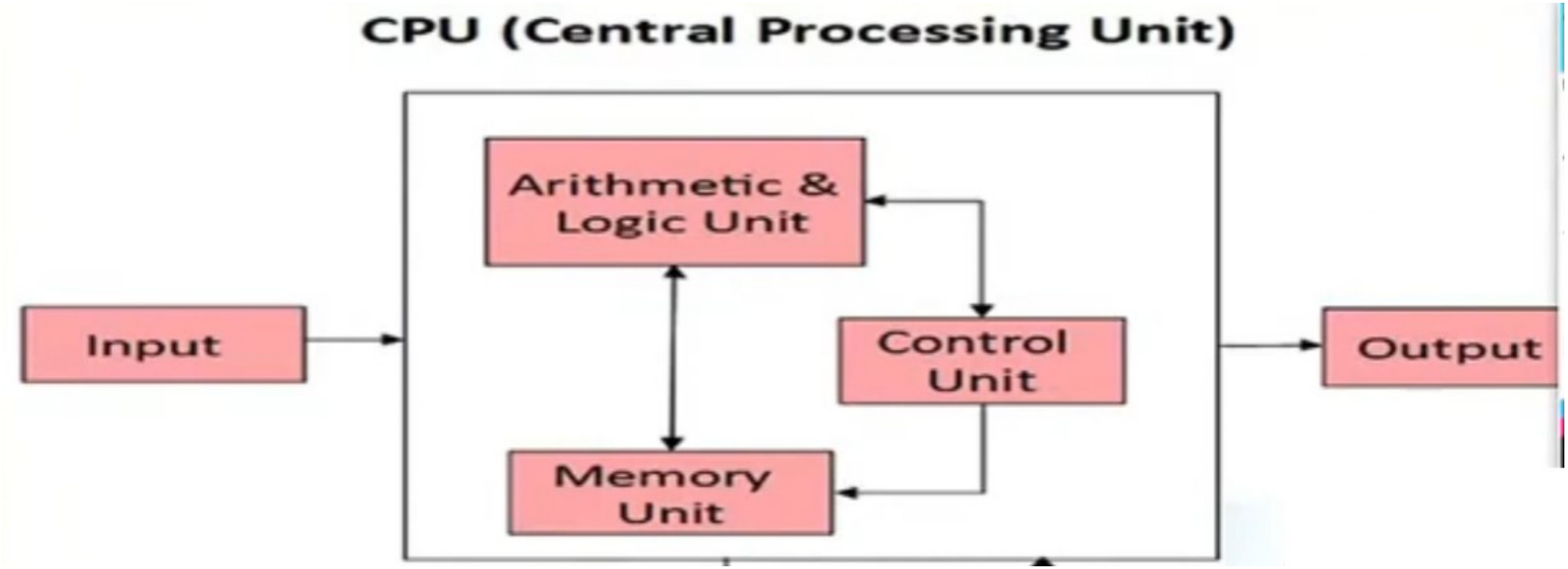
3 Cache size

Bigger cache means fewer slow trips to RAM.

4 Architecture and bits

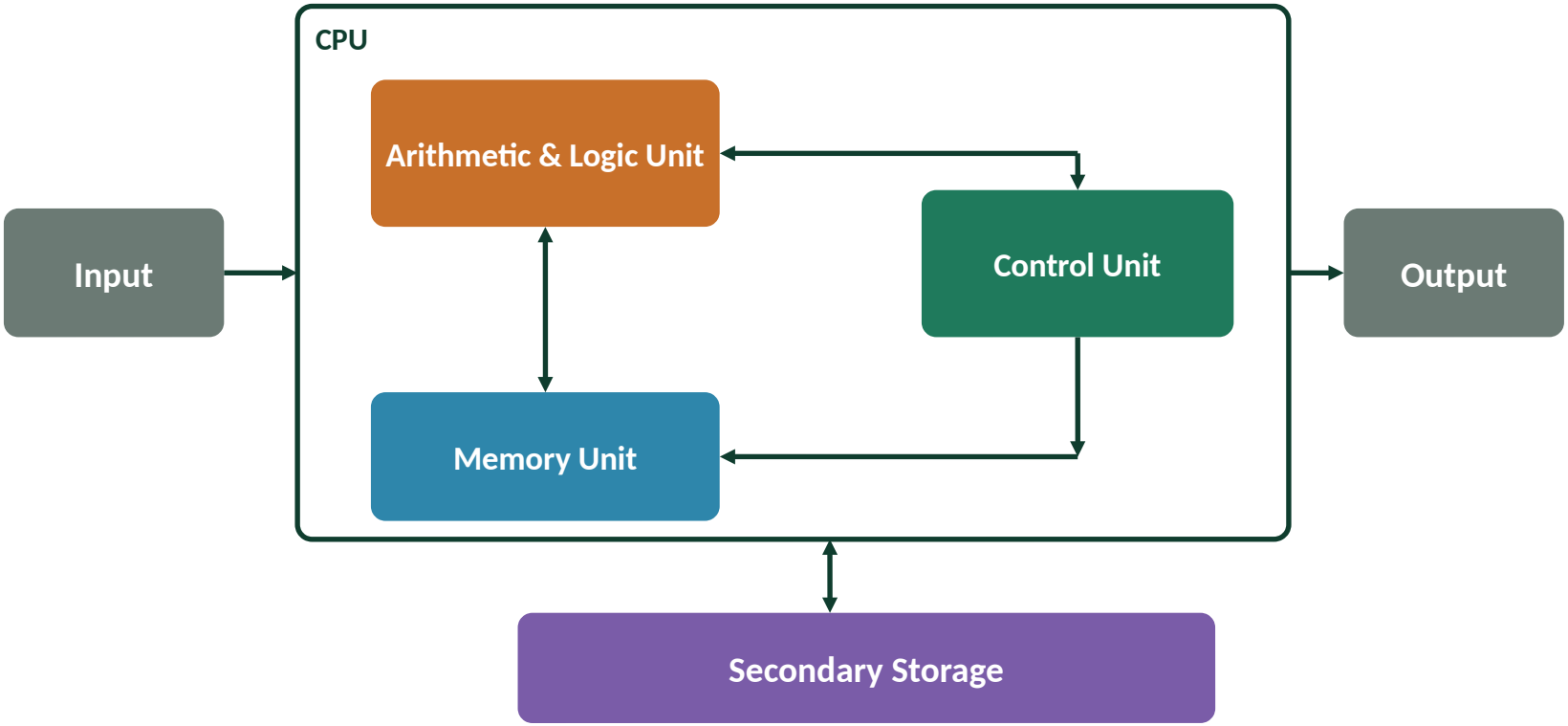
A newer, wider design does more per cycle.

Functions of CPU



Lecture diagram: Input → CPU (Arithmetic & Logic Unit, Control Unit, Memory Unit) → Output. Secondary Storage is added in the next slide (cut off in the screenshot).

CPU block diagram – Explained



Functions – Explained

1 Input

Data and instructions enter the system.

2 Process

ALU calculates, CU directs and the memory unit holds data.

3 Output

The result goes to the screen, speaker or printer.

4 Storage

Secondary storage keeps data permanently; the CPU loads it when needed.

How the CPU works

FROM THE LECTURE

- The processor first fetches and instruction from the meain memory
- The instruction is then decoded to determine what action is required to be done.
- Based on instruction the processor fetches, if required, data from main memory or i/o module.
- The instruction is then executed which may require performing arithmetic or logical operations on data.
- In addition to execution, CPU also supervises and controls i/o devices. It there is any request from i/o devices, called interrupt, the CPU Suspends Handling Program.
- Finally the results of an execution may required transfer of data to … module

Machine cycle



1 Fetch

Get the instruction from main memory (RAM).

2 Decode

The control unit works out what to do.

3 Execute

The ALU does the arithmetic or logic.

4 Store

The result goes to a register, RAM or an I/O module.

Interrupts

What is an interrupt?

A request from an I/O device (keyboard, mouse, disk) asking for the CPU's attention.

What the CPU does

It suspends the current program, handles the request, then returns.

Why it matters

The CPU need not keep checking devices, so the computer feels responsive.

Example

You press a key: the keyboard interrupts the CPU, which reads the key and continues.

Types of microprocessor

1 By bits (lecture)

32 BIT and 64 BIT architecture. Earlier CPUs were 4, 8 and 16 bit.

2 By instruction set

CISC (many complex instructions) and RISC (few simple, fast instructions).

3 By purpose

General-purpose CPUs for PCs, embedded chips for devices, and graphics processors (GPU).

4 By brand

Intel and AMD for PCs; ARM-based chips for phones.

CISC vs RISC

	CISC	RISC
Full form	Complex Instruction Set Computer	Reduced Instruction Set Computer
Instructions	Many, complex, variable length	Few, simple, fixed length
Speed per instruction	May take several cycles	Usually one cycle
Examples	Intel x86 family	ARM; Intel/AMD translate into RISC internally
Link to lecture	Older designs	Third generation (RISC)

History of CPU

INTEL’S 1st Processor

Intel's 1st Processor was INTEL 4004 which was designed for Calculators

AMD’S 1st Processor

AMD's 1st Processor was AMD AM9300 which was designed as Competitor of INTEL

INTEL’S Latest Processor

Intel's latest Processor is INTEL i9 10TH Generation (Socket LGA 2066)

AMD’S Latest Processor

AMD's latest Processor is RYZEN 9 3000+ Series (Socket AM4)

History of CPU – Explained

1 Intel 4004

Released in 1971, the first commercial microprocessor: a 4-bit chip made for calculators.

2 AMD

Started as a competitor to Intel and has been its main rival in PC processors since.

3 Latest at that time

Intel i9 10th generation and Ryzen 9 3000+ were the newest when the lecture was made.

4 Today

Newer generations keep appearing with new sockets (for example AMD AM5), so always check the current model.

Microprocessor Generations

First generation: 1971-78

Behind the power curve (16-bit, <50k transistors)

Second generation: 1979-85

Becoming “real” computers (32-bit , >50k transistors)

Third generation: 1985-89

Challenging the “establishment” (Reduced Instruction Set Computer/RISC, >100k transistors)

Fourth generation: 1990-

Architectural and performance leadership (64-bit, > 1M transistors, Intel/AMD translate into RISC internally)

Generations – Timeline

1st gen	2nd gen	3rd gen	4th gen
1971-78	**1979-85**	**1985-89**	**1990-**
<50k transistors	>50k transistors	>100k transistors	>1M transistors
16-bit	**32-bit**	**RISC**	**64-bit**

1 **Pattern**

Each generation has more transistors, wider data (16 → 32 → 64 bit) and higher speed.

Generations – Explained

1 First

Simple chips that were weaker than the big computers of the time.

2 Second

32-bit chips with more power brought real PCs.

3 Third

RISC designs challenged the old designs and raised speed.

4 Fourth

64-bit, millions of transistors, multi-level cache and later multiple cores.

Components of CPU

1 **ALU**

Arithmetic & Logic Unit: does calculations and comparisons.

2 **CU**

Control Unit: directs and times every operation.

3 **MU**

Memory Unit: registers and internal memory close to the CPU.

4 **Cache Memory**

Very fast memory that keeps frequently used data.

ALU – Arithmetic & Logic Unit

1 Arithmetic

Addition, subtraction, multiplication and division.

2 Logic

AND, OR, NOT and comparisons such as greater than, less than, equal.

3 Works with registers

Takes data from registers, computes, and puts the result back.

4 Flags

Records the outcome (zero, negative, carry) so the CU can decide what next.

CU – Control Unit

1 Decodes

Reads the instruction register and understands the instruction.

2 Controls

Sends control signals to ALU, memory and I/O devices.

3 Times

Keeps all parts in step with the clock.

4 Directs data flow

Decides which data moves where, using the buses.

MU – Memory Unit

1 Registers

Tiny, ultra-fast storage inside the CPU holding the data being used right now.

2 Internal memory

Holds instructions and data close to the ALU and CU.

3 Link to RAM

Main memory (RAM) feeds the memory unit.

4 Volatile

Contents are lost when the power is off.

Cache memory

1 What it is

A small, very fast memory between the CPU and RAM.

2 Why

RAM is slow compared with the CPU. Cache keeps copies of frequently used data.

3 Hit and miss

Data found in cache = hit (fast). Not found = miss, read from RAM.

4 Levels

L1 (smallest, fastest), L2, and L3 (larger, shared by cores).

Memory hierarchy

Level	Speed	Typical size
Registers	Fastest	Bytes
L1 cache	Very fast	Tens of KB per core
L2 cache	Fast	Hundreds of KB to a few MB
L3 cache	Fast, shared	Several MB and more
RAM	Slower	GBs
HDD / SSD	Slowest	Hundreds of GB to TB

Register Set (IR, AR, MA)

1 **IR**

Instruction Register: holds the instruction that is being executed now.

2 **AR**

Address Register: holds the address of the data or instruction.

3 **MA**

Memory Address (register): the address sent to memory to read or write.

4 **Others**

Program Counter (next instruction), Accumulator (ALU result) and flag register.

Registers in the cycle



1 Fetch

The address of the next instruction goes to memory.

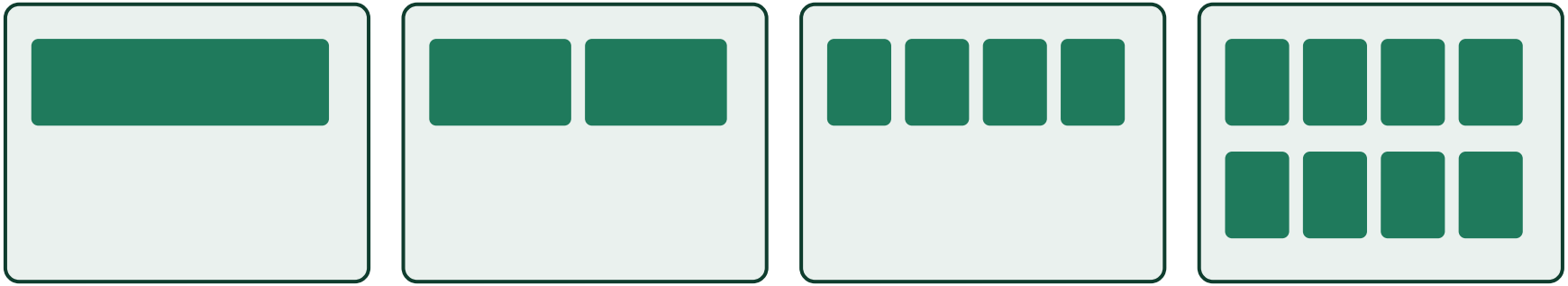
2 Decode

The instruction sits in IR while the CU decodes it.

3 Execute

Data in registers goes to the ALU; the result is stored back.

Core technologies



Single core | Dual core | Quad core | Octa core

1 Core

A core is a complete processing unit with its own ALU, CU and registers.

2 Multi-core

Many cores in one chip handle many tasks at once.

3 Benefit

Smoother multitasking; heavy programs can split work across cores.

Core technologies – Explained

1 Dual core

2 cores: good for basic work.

2 Quad core

4 cores: common for home and office.

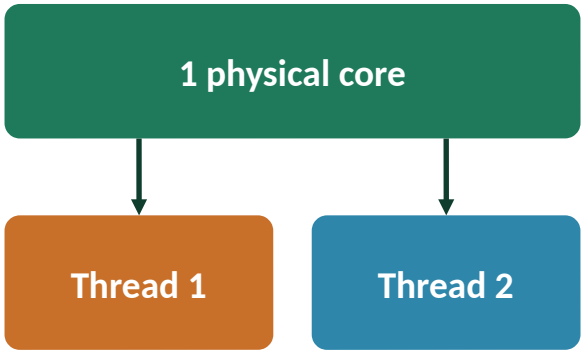
3 Hexa and octa

6 or 8 cores: gaming and editing.

4 Limits

Programs must be written to use many cores; more cores use more power and heat.

Hyper threading



Intel technology
Hyper-Threading makes one physical core appear as two logical processors.

How
While one thread waits, the core runs the other, so idle time is used.

Result
Example: 4 cores / 8 threads. Up to a good boost, but not double the speed.

Hyper threading – Explained

	Without HT	With HT
Quad core	4 cores, 4 threads	4 cores, 8 threads
Idle time	Wasted while waiting	Used by the second thread
Best for	Single heavy tasks	Many tasks and multitasking
Visible in	Task Manager	Logical processors count

Internal Buses

1 **Data bus**

Two-way: carries data between CPU, memory and devices.

2 **Address bus**

One-way: carries the memory address the CPU wants to read or write.

3 **Control bus**

Carries control signals such as read, write and clock.

4 **Internal vs external**

Internal buses connect parts inside the CPU; external ones connect the CPU to the motherboard.

Bus width

Bus	Direction	Width tells you
Data bus	Both ways	How many bits move at once (32 or 64)
Address bus	CPU to memory	Memory reachable: 2^n locations
Control bus	Both ways	Read/write, interrupt and clock signals

Example: a 32-bit address bus can reach 2^32 locations = about 4 GB. This is why 32-bit systems are limited to about 4 GB of RAM.

Sockets

FROM THE LECTURE

- According to Structure Sockets are a collection or a Array of Pins which holds Microprocessor.
- Sockets are used to adjust the position of Microprocessor in the Motherboard. These are Categorized according to the Brands and Version of Microprocessors…

Socket types

INTEL	No. of pins	Processor
LGA	1151 to 1155	I3, i5 & i7
PGA	Text	Pentium
PPGA	Text	Text

AMD	No. of pins	Processor
C Series	Socket C32	Opteron
AM Series	AM 1 to 4	Ryzen 3, 5, 7, Phenom,
A Series	Socket A462	…

Table copied from the lecture slide (“Text” and “…” are placeholders or hidden parts in the screenshot).

Sockets – Explained

1 LGA

Land Grid Array: pins are in the motherboard socket; the CPU has flat contacts. Used by Intel.

2 PGA

Pin Grid Array: pins are on the CPU and fit into holes in the socket (older Intel, AMD).

3 AM series

AMD desktop sockets (AM1 to AM4 and newer), used for Ryzen, Phenom and others.

4 Compatibility

A CPU fits only its matching socket, so match CPU and motherboard before buying.

Sockets – Examples

Socket	Used for
LGA 1155 / 1151	Intel Core i3, i5, i7 (as in lecture)
LGA 2066	High-end Intel i9 (X series)
AM4	AMD Ryzen 3000 series
C32	AMD Opteron server CPUs
A462	Older AMD Athlon / Duron

Summary

1 CPU

Brain of the computer; chip of transistors and logic gates; 32 or 64 BIT.

2 Components

ALU calculates, CU controls, MU and cache hold data, registers (IR, AR, MA) store instructions.

3 Performance

Clock speed, cores, hyper-threading, cache and bus width decide speed.

4 Platform

Generations improved transistors and bits; sockets connect the CPU to the motherboard.

Thank you

CPU = fetch, decode, execute – billions of times every second.